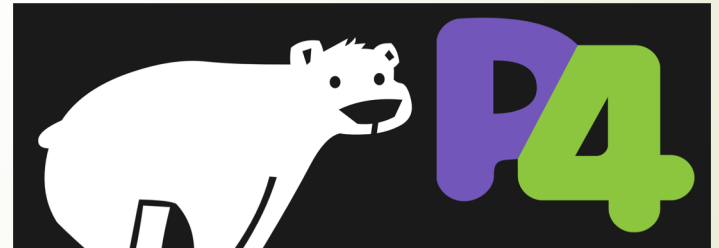


CompSci 234/NetSys 210  
Advanced Topics in Networking  
**Winter 2019**

# Unit 11: Dataplan Programming with P4



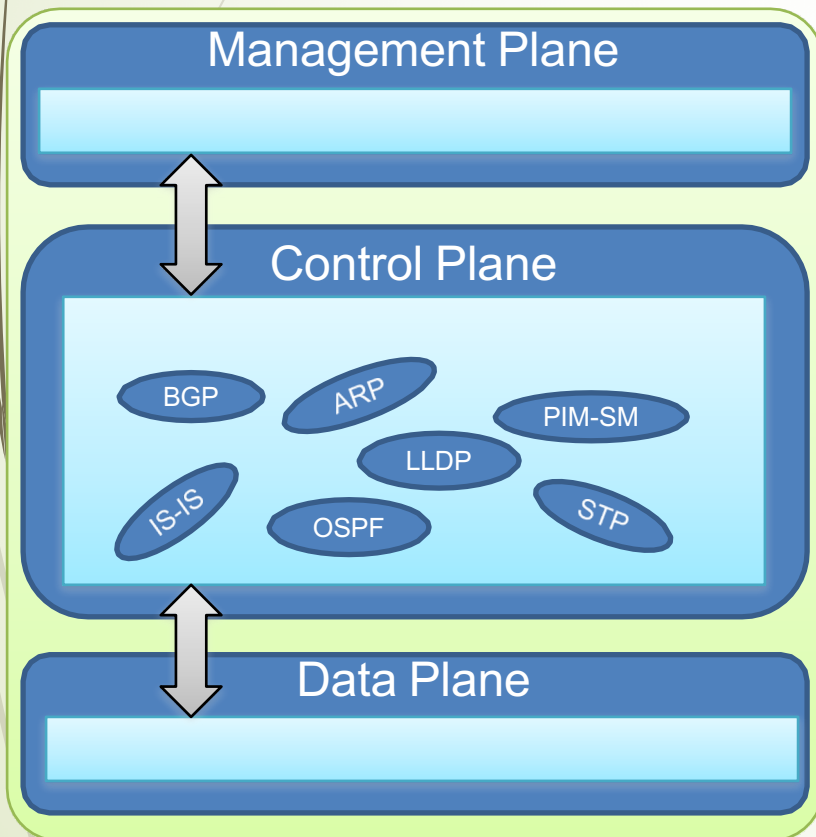
Cheng-Hsin Hsu ([chsu@cs.nthu.edu.tw](mailto:chsu@cs.nthu.edu.tw))

Slide adopted from p4.org materials, in particular, from V. Gurevich's tutorial

# Agenda

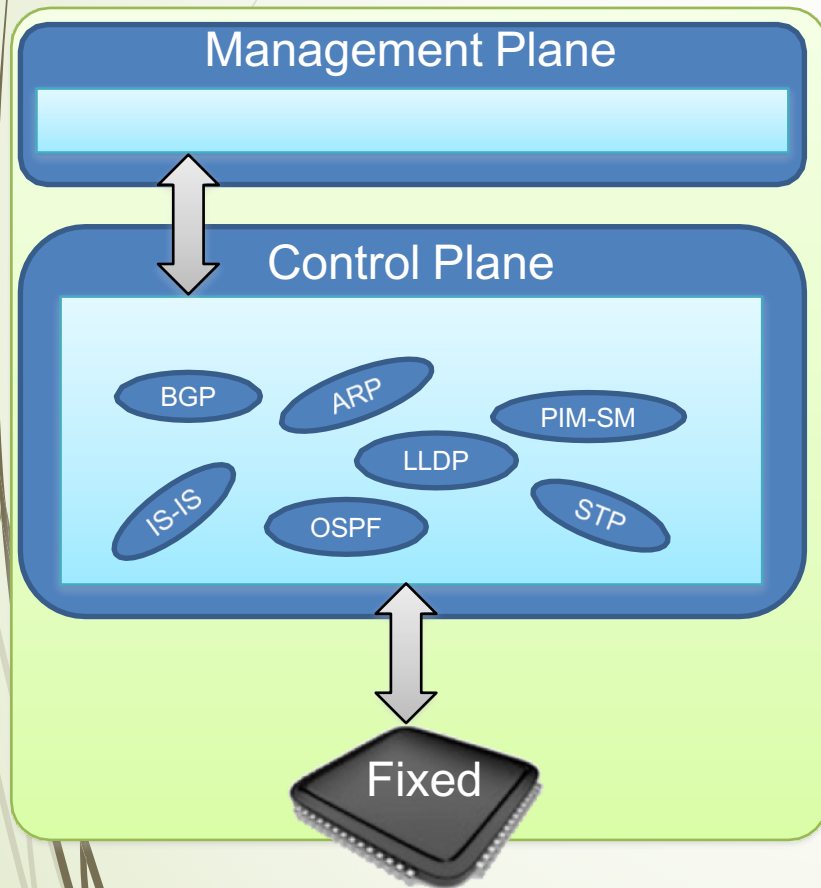
- Why Dataplan Programming
- P4 Overview
- Main Data Types
- Packet Parsing
- Packet Processing
- Packet Deparsing
- Program Structure
- P4 Runtime
- Summary

# Standard Telecommunications Architecture



- **Traditional architecture consists of the three planes**
- **A Plane is a group of algorithms**
- **These algorithms**
  - Process different kinds of traffic
  - Have different performance requirements
  - Are designed using different methodologies
  - Are implemented using different programming languages
  - Run on different hardware

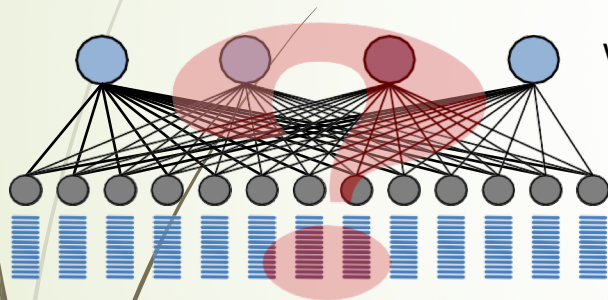
# What Constitutes a NOS?



- **Control Plane**
- **Management Plane**
- **How about Data Plane?**
  - In many modern, high-speed devices it is **no longer** a part of a NOS

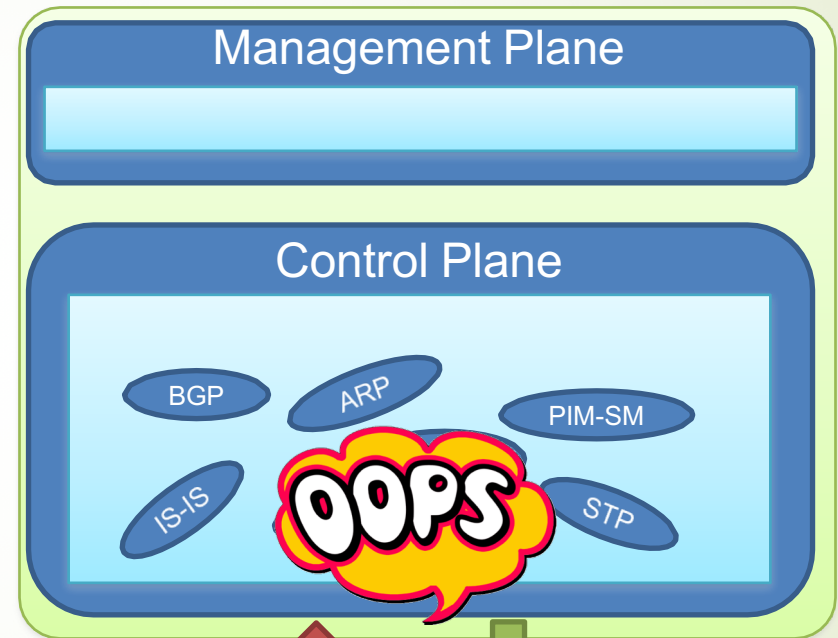


# Designing NOS for a Fixed Data Plane

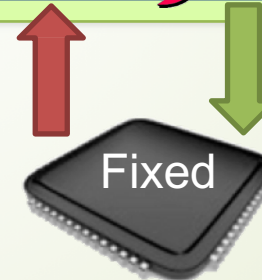


That's how I  
want to build my  
network!

Requirements

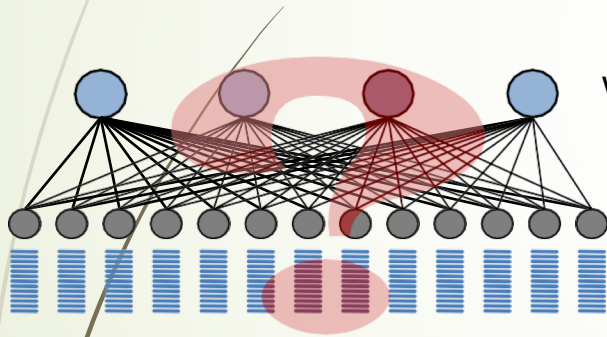


That's how I  
process the  
packets!



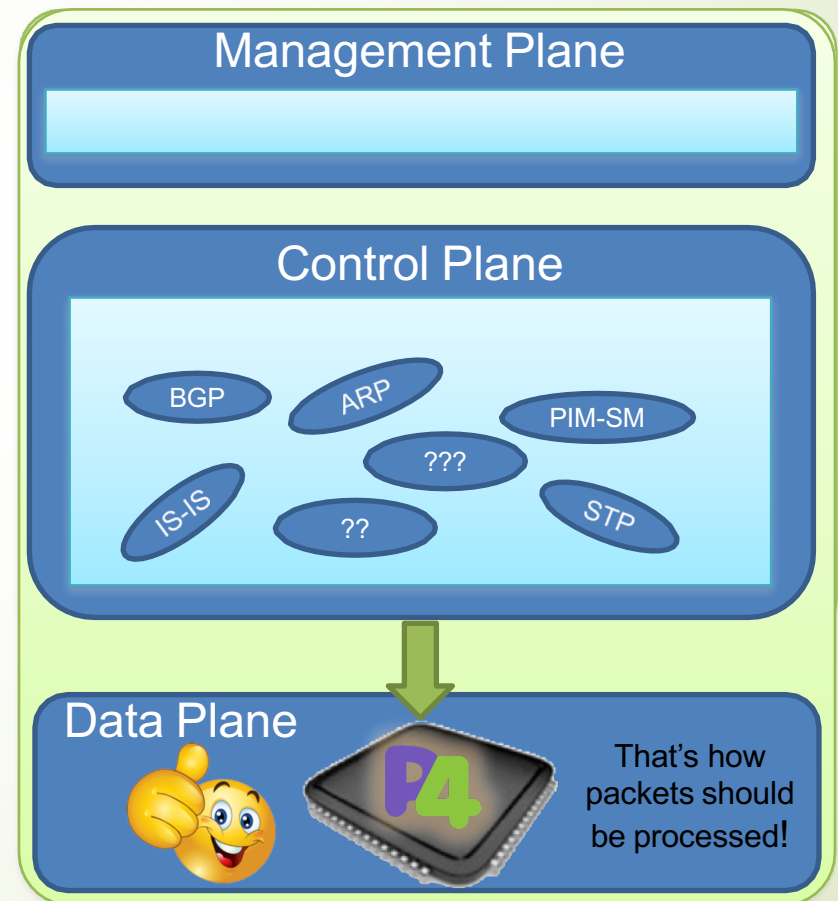
That's how  
packets should  
be processed!

# Bringing Data Plane back in the NOS



That's how I  
want to build my  
network!

Requirements



# Advantages of Data Plane Programmability

C

- Control and Customization. Make the device behave exactly as you want

R

- Reliability. Reduce the risk by removing unused features

E

- Efficiency. Reduce energy consumption and expand scale by doing only what you need

A

- Add new features on your schedule/roadmap

T

- Telemetry. Be able to see inside the Data Plane

E

- Exclusivity and Differentiation. No need to share your IP with the chip vendor



# Agenda

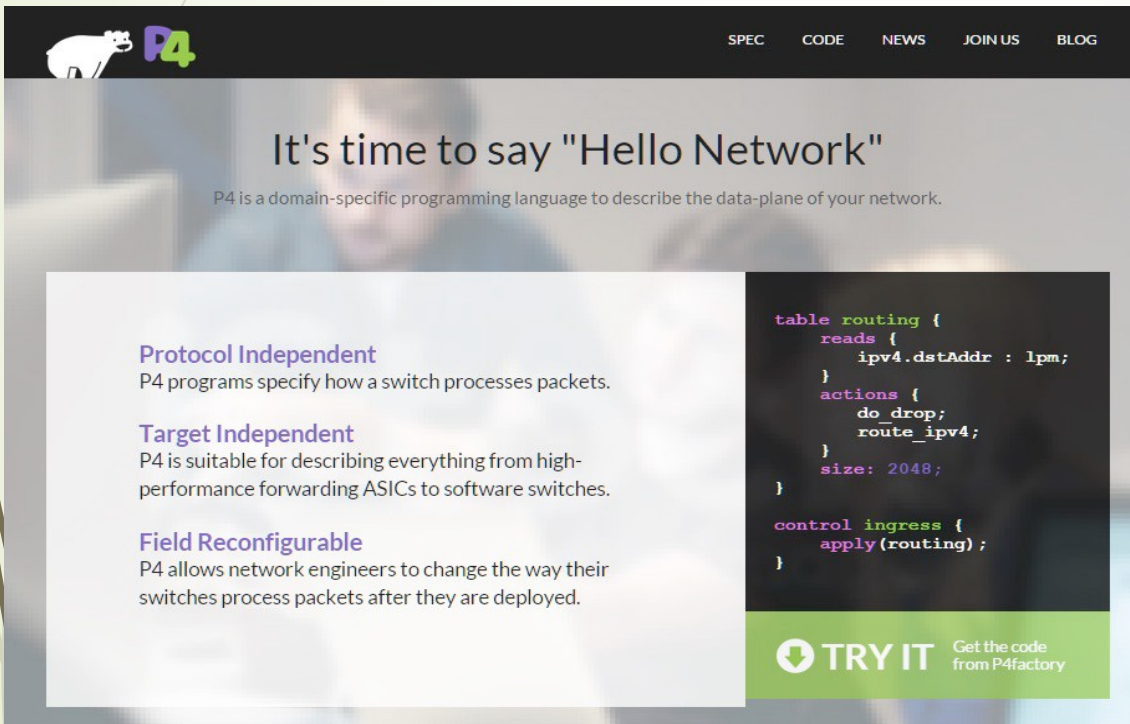
- Why Dataplan Programming
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# Brief History and Trivia

- **May 2013:** Initial idea and the name “P4”
- **July 2014:** First paper (SIGCOMM ACR)
- **Aug 2014:** First P4<sub>14</sub> Draft Specification (v0.9.8)
- **Sep 2014:** P4<sub>14</sub> Specification released (v1.0.0)
- **Jan 2015:** P4<sub>14</sub> v1.0.1
- **Mar 2015:** P4<sub>14</sub> v1.0.2
- **Nov 2016:** P4<sub>14</sub> v1.0.3
- **May 2017:** P4<sub>14</sub> v1.0.4
  
- **Apr 2016:** P4<sub>16</sub> – first commits
- **Dec 2016:** First P4<sub>16</sub> Draft Specification
- **May 2017:** P4<sub>16</sub> Specification released
- . . .
  
- **Official Spelling** P4\_16 on terminals, P4<sub>16</sub> in publications

# The P4 Language Consortium



The screenshot shows the P4 Consortium website. At the top is a navigation bar with the P4 logo (a bear and the text 'P4') on the left and links for 'SPEC', 'CODE', 'NEWS', 'JOIN US', and 'BLOG' on the right. The main heading is 'It's time to say "Hello Network"', followed by the subtext 'P4 is a domain-specific programming language to describe the data-plane of your network.' Below this, there are three columns of text. The first column lists three key features: 'Protocol Independent', 'Target Independent', and 'Field Reconfigurable', each with a brief description. The second column contains a code snippet in P4 language. The third column has a green button labeled 'TRY IT' with a download icon and the text 'Get the code from P4factory'.

**Protocol Independent**  
P4 programs specify how a switch processes packets.

**Target Independent**  
P4 is suitable for describing everything from high-performance forwarding ASICs to software switches.

**Field Reconfigurable**  
P4 allows network engineers to change the way their switches process packets after they are deployed.

```
table routing {
  reads {
    ipv4.dstAddr : lpm;
  }
  actions {
    do_drop;
    route_ipv4;
  }
  size: 2048;
}

control ingress {
  apply(routing);
}
```

 **TRY IT** Get the code from P4factory

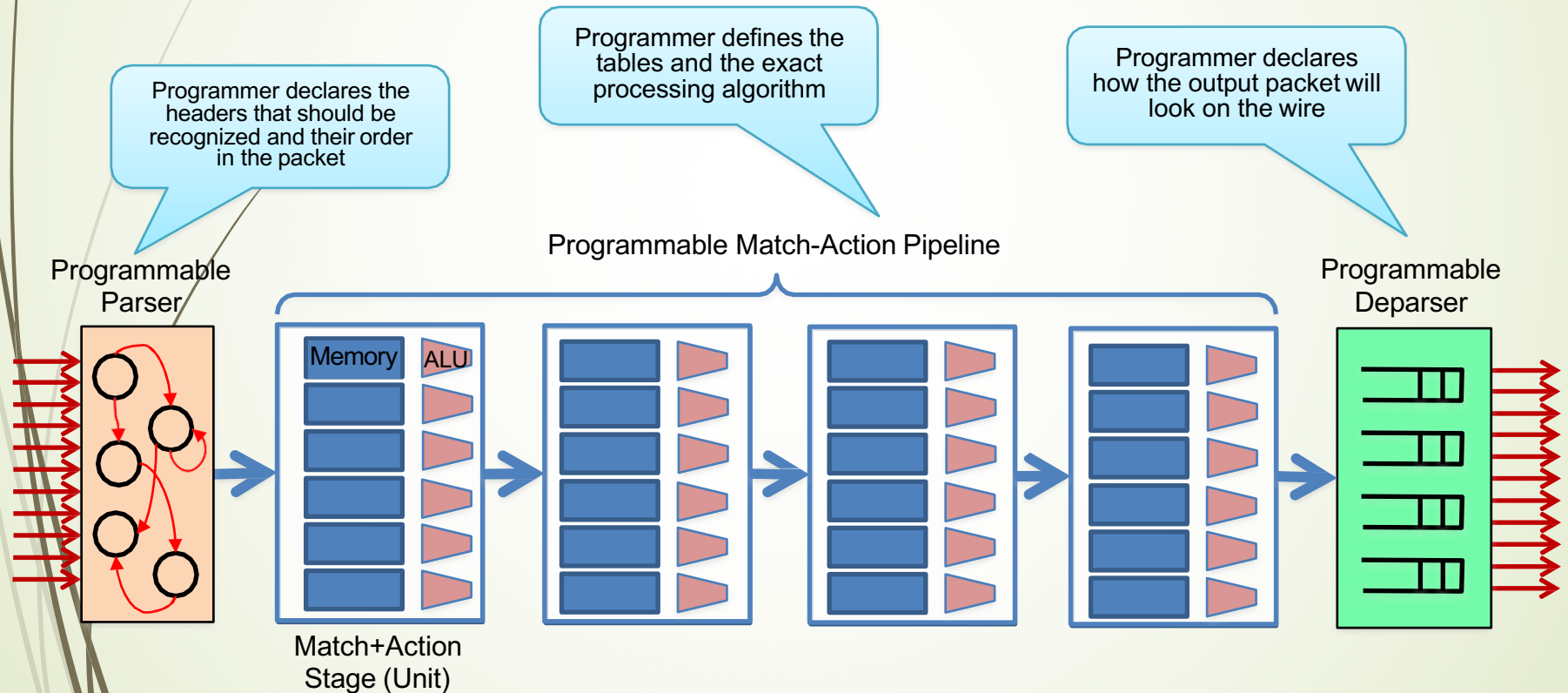
- Consortium of academic and industry members
- Open source, evolving, domain-specific language
- Permissive Apache license, code on GitHub today
- Membership is free: contributions are welcome
- Independent, set up as a California nonprofit

# P4<sub>16</sub> Design Goals

- **Logical Evolution of P4<sub>14</sub>**
  - Same basic building blocks and concepts
  - More expressive and convenient to use
  - More formally defined semantics
    - Strong Type System
  - Support for good software engineering practices
- **Target-Independence**
  - Support for a variety of targets (ASICs, FPGAs, NICs, software)
    - Language/Architecture separation
    - Flexible data plane model
- **Non-goals**
  - New constructs
  - General-purpose programming

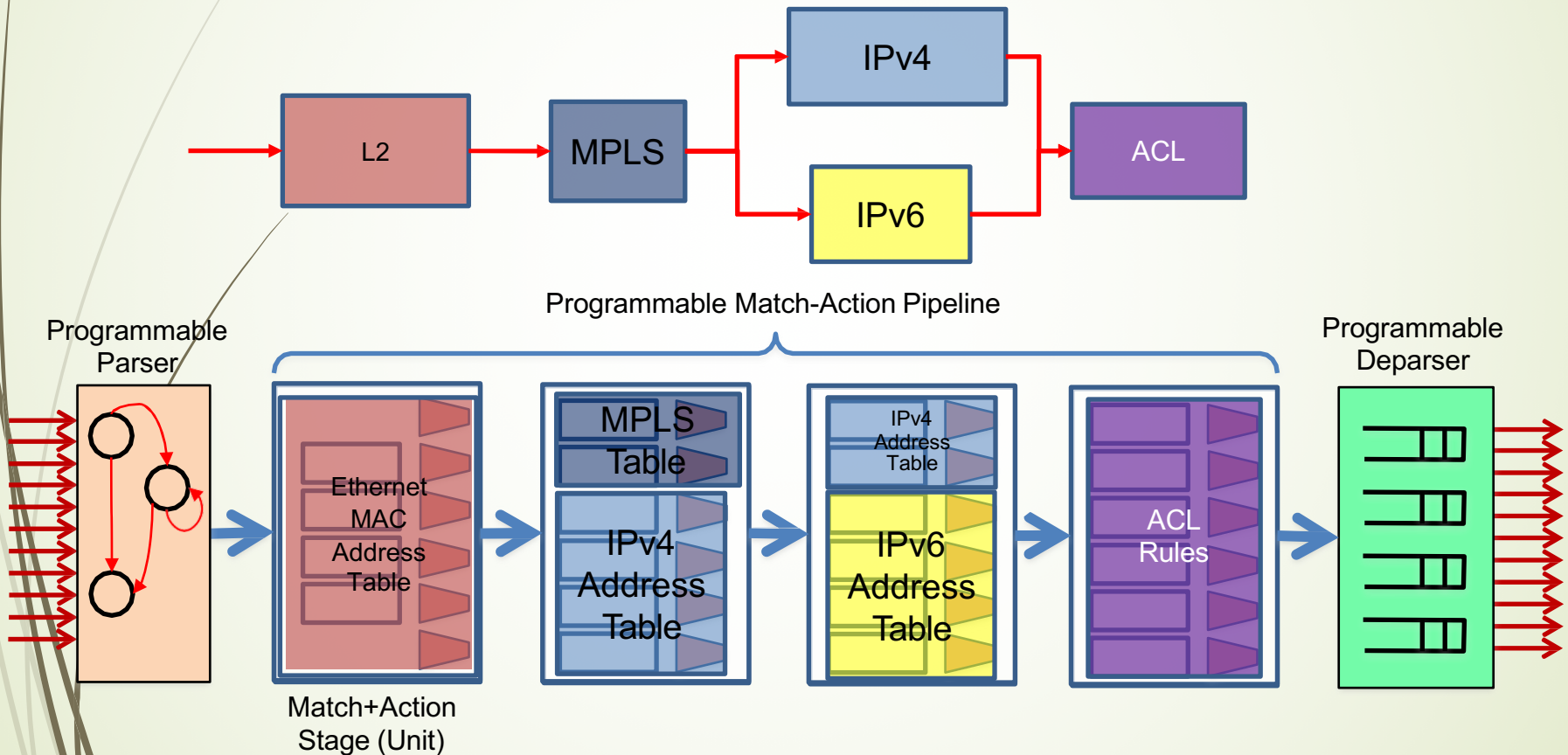


# PISA: Protocol-Independent Switch Architecture





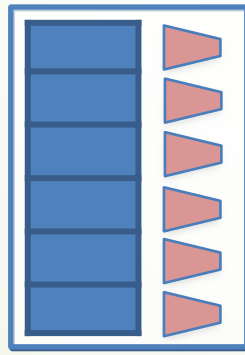
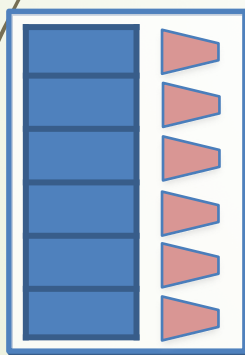
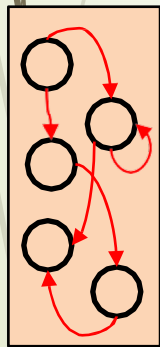
# Sample Data Plane Program on PISA



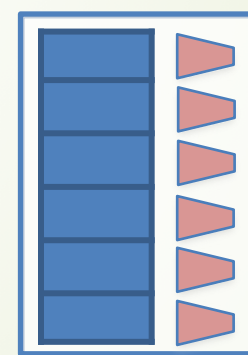
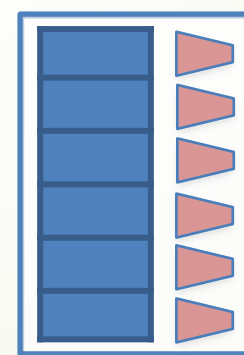
# P4<sub>14</sub> Switch Model

- **Ingress Pipeline**
- **Egress Pipeline**
- **Traffic Manager**
  - N:1 Relationships: Queueing, Congestion Control
  - 1:N Relationships: Replication
  - Scheduling

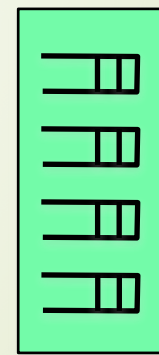
Programmable  
Parser



Packet  
Queueing,  
Replication &  
Scheduling



Implicitly  
Programmable  
Deparser



Ingress pipeline

Egress pipeline